

JPHG02 Heat and internal energy — Quiz 1

S3 Physics · 4 multiple-choice items

Q1 · JPHG02.2 · Standard

An 800 W electric heater is switched on for 3 minutes. How much electrical energy does it use? ($E = Pt$)

- **A.** 2,400 J
- **B.** 48,000 J
- **C.** 144,000 J
- **D.** 1,440,000 J

Q2 · JPHG02.3 · Standard

A 0.30 kg sample of water is heated from 25°C to 65°C. How much energy does it gain? ($E = mc\Delta T$)

Specific heat capacity of water = 4,200 J/(kg°C)

- **A.** 5,040 J
- **B.** 12,600 J
- **C.** 50,400 J
- **D.** 504,000 J

Q3 · JPHG02.4 · Standard

A 0.40 kg metal block absorbs 18,000 J of energy. Its temperature rises by 50°C. What is its specific heat capacity? Assume no energy is lost.

- **A.** 90 J/(kg°C)
- **B.** 360 J/(kg°C)
- **C.** 900 J/(kg°C)
- **D.** 2,250 J/(kg°C)

Q4 · JPHG02.6 · Standard

A 0.28 kg aluminium block at 100°C is placed in 0.18 kg of water at 20°C in an insulated container. What is their final temperature? Ignore energy absorbed by the container.

Specific heat capacity of aluminium = 900 J/(kg°C)

Specific heat capacity of water = 4,200 J/(kg°C)

- **A.** 30°C
- **B.** 40°C
- **C.** 60°C
- **D.** 80°C

JPHG02 Heat and internal energy — Quiz 1 · Answer key

Teacher page · 4 items

Q1 · JPHG02.2

Answer: C. 144,000 J

Guide: Time is $3 \times 60 = 180$ s. $E = Pt = 800 \text{ W} \times 180 \text{ s} = 144,000 \text{ J}$.

Q2 · JPHG02.3

Answer: C. 50,400 J

Guide: $\Delta T = 65^\circ\text{C} - 25^\circ\text{C} = 40^\circ\text{C}$. $E = mc\Delta T = 0.30 \text{ kg} \times 4,200 \text{ J}/(\text{kg}^\circ\text{C}) \times 40^\circ\text{C} = 50,400 \text{ J}$.

Q3 · JPHG02.4

Answer: C. 900 J/(kg°C)

Guide: $c = E / (m\Delta T) = 18,000 \text{ J} / (0.40 \text{ kg} \times 50^\circ\text{C}) = 900 \text{ J}/(\text{kg}^\circ\text{C})$.

Q4 · JPHG02.6

Answer: B. 40°C

Guide: Energy lost by the aluminium equals energy gained by the water: $0.28 \times 900 \times (100 - T) = 0.18 \times 4,200 \times (T - 20)$. This gives $252(100 - T) = 756(T - 20)$, so $T = 40^\circ\text{C}$.